Sickle cell anaemia

☞ **1** Circle the correct word in each pair.

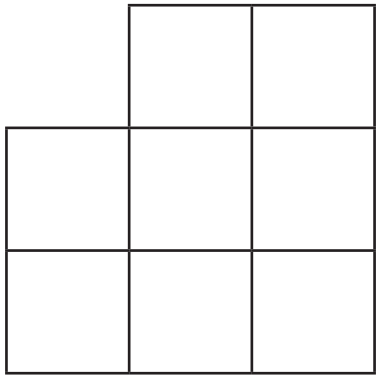
Many **genetic mutations** can be harmful to an organism/organizm but some can also be benificial/beneficial. In Africa a generic/genetic mutation known as sickle-cell trait has been found to provide resistance to malaria. Malaria is a serious disese/disease which kills millions of people/person every year. It is caused by a microscopic/microscope parasite which attacks read/red blood cells. People with sickle-cell trait have an increased resistance to milaria/malaria because their blood/bleed cells are oddly shaped and won't host the parasite/parasyte properly. Resistent/Resistant people are **heterozygous**, which means they carry only one **allele** of the sickle-cell trait. Those who are **homozygous** have two alleles/allelles of the sickle-cell trait. These individuals are weakened and have a short/shirt life-span. This condition is called sickle cell anaemia.

2 A sickle-celled red blood cell has a flattened shape and can't pick up **oxygen** as efficiently as a normal red blood cell. How would this create health problems for an individual?

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3 The allele for sickle-cell trait is Hb^S and for regular blood cells it is Hb^A. Fill out the punnet square below showing a cross between a homozygous person (2 genes the same) and a heterozygous person (1 gene different to the other) for the sickle-cell allele.



4 What are the resulting **genotypes** and **phenotypes**? _____

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